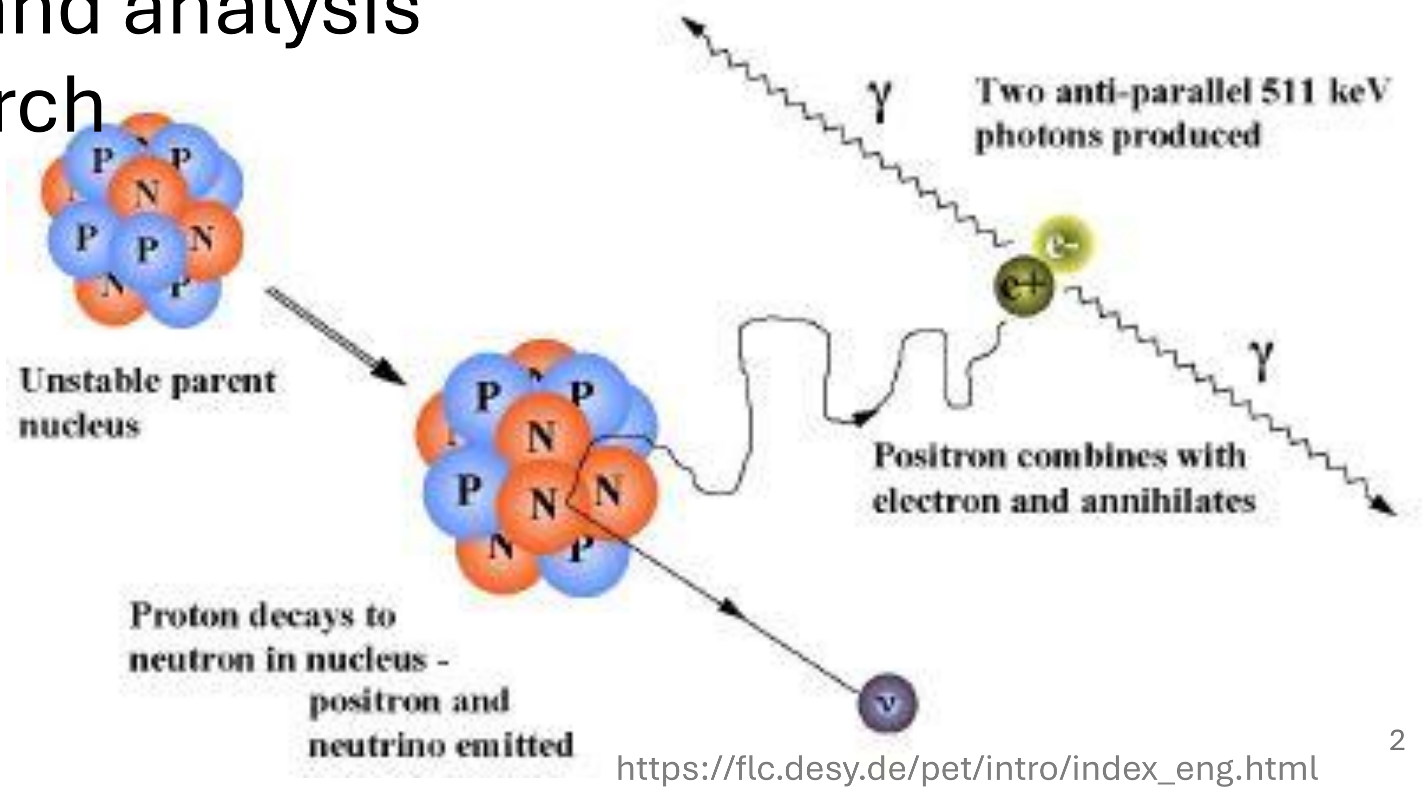
Impact of positron range on image reconstruction in ^{18}F and ^{44}Sc PET imaging

12.03.2026

Anna Giebułtowska

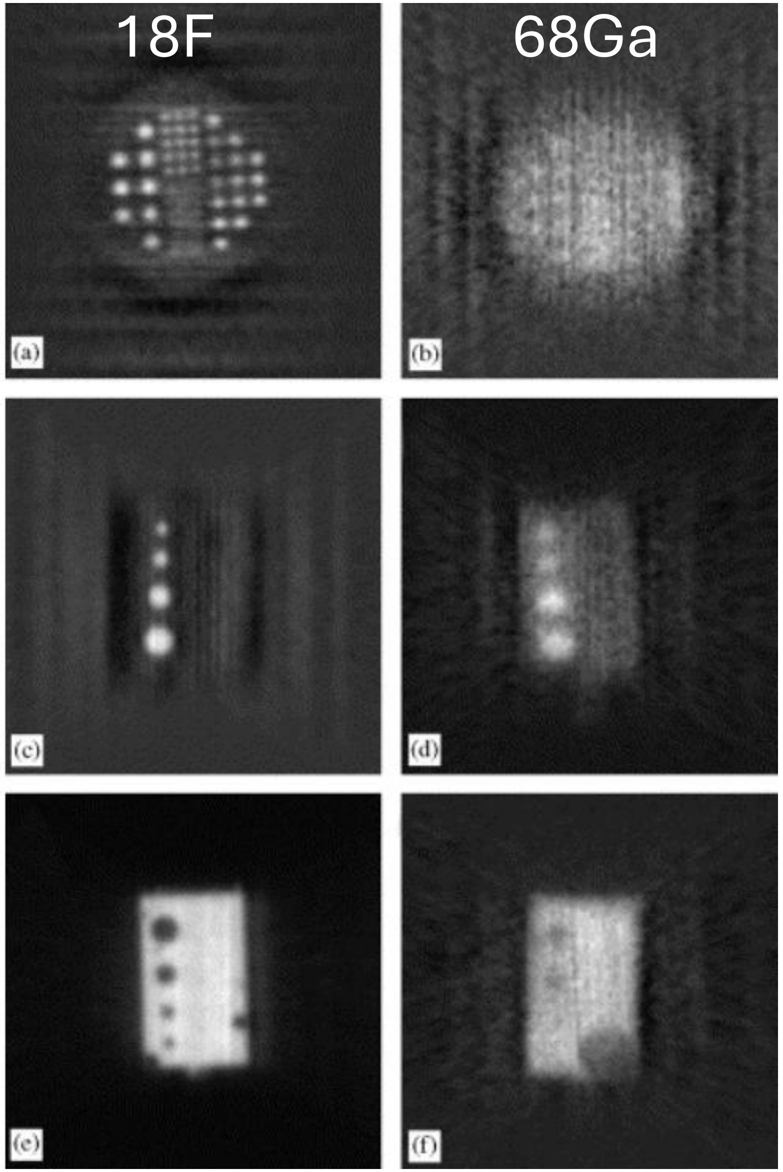
Agenda

- Motivation
- Scandium literature value found
- Computing ranges for different isotopes
- Experiment and analysis
- Future research



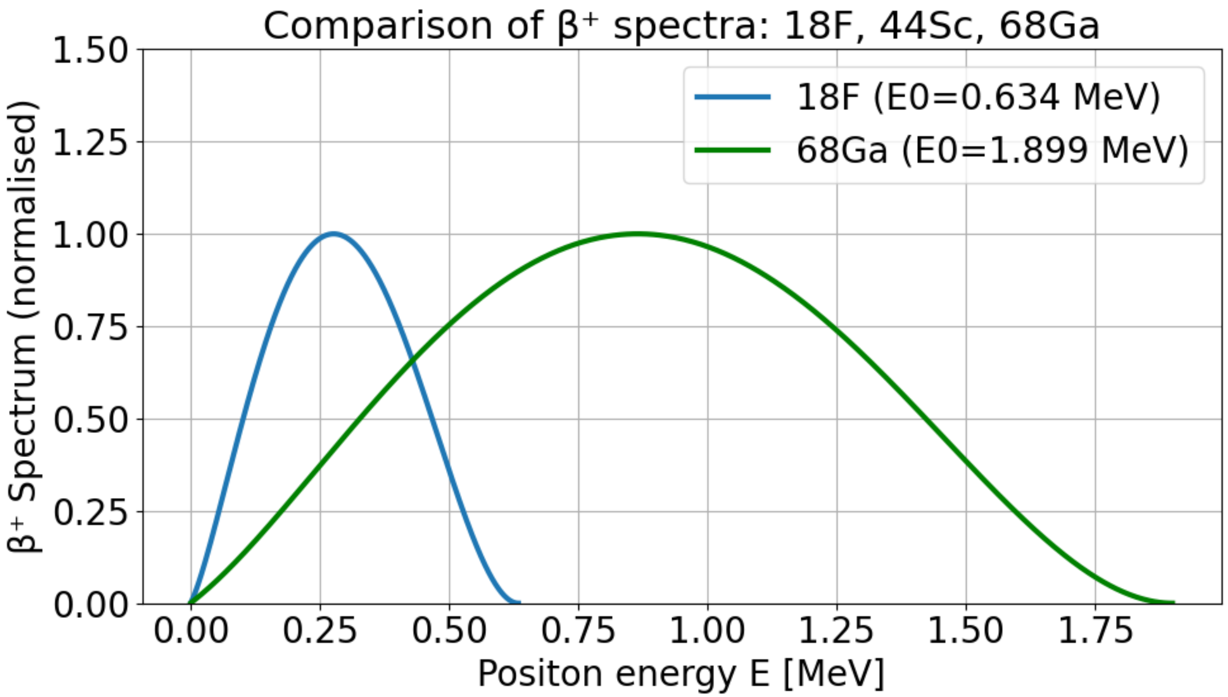
https://flc.desy.de/pet/intro/index_eng.html

Motivation



Partige: https://www.sciencedirect.com/science/article/pii/S0168900206016640

Levin CS, Hoffman EJ. Calculation of positron range and its effect on PET resolution. Phys Med Biol.



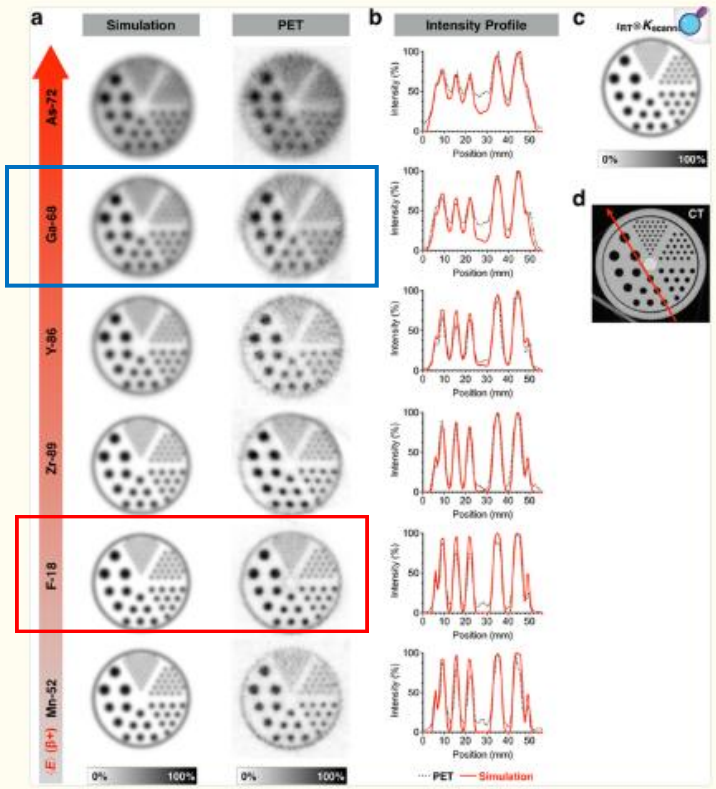
$$N(E)\,dE = g\,F(Z,E)\,pE(E_{\max}-E)^2\,dE$$

$$F_{\text{allowed}}(Z,E) = \frac{2\pi\eta}{1-e^{-2\pi\eta}}$$

$$\eta = -Z\alpha E/p$$

Motivation

Fig. 5.



Open in a new tab

a Comparison of PET images and simulations of the Data Spectrum® Micro Jaszczak phantom with different positron emitters. Data are normalized to the maximum pixel intensity in each respective image. b Intensity profiles taken over line indicated in d. c Simulated distribution of radiotracer (i.e., zero positron range blur) convolved with $PSF_{scanner}$ represents upper limit on achievable resolution. d Computed tomography (CT) scan of the unfilled Jaszczak phantom.

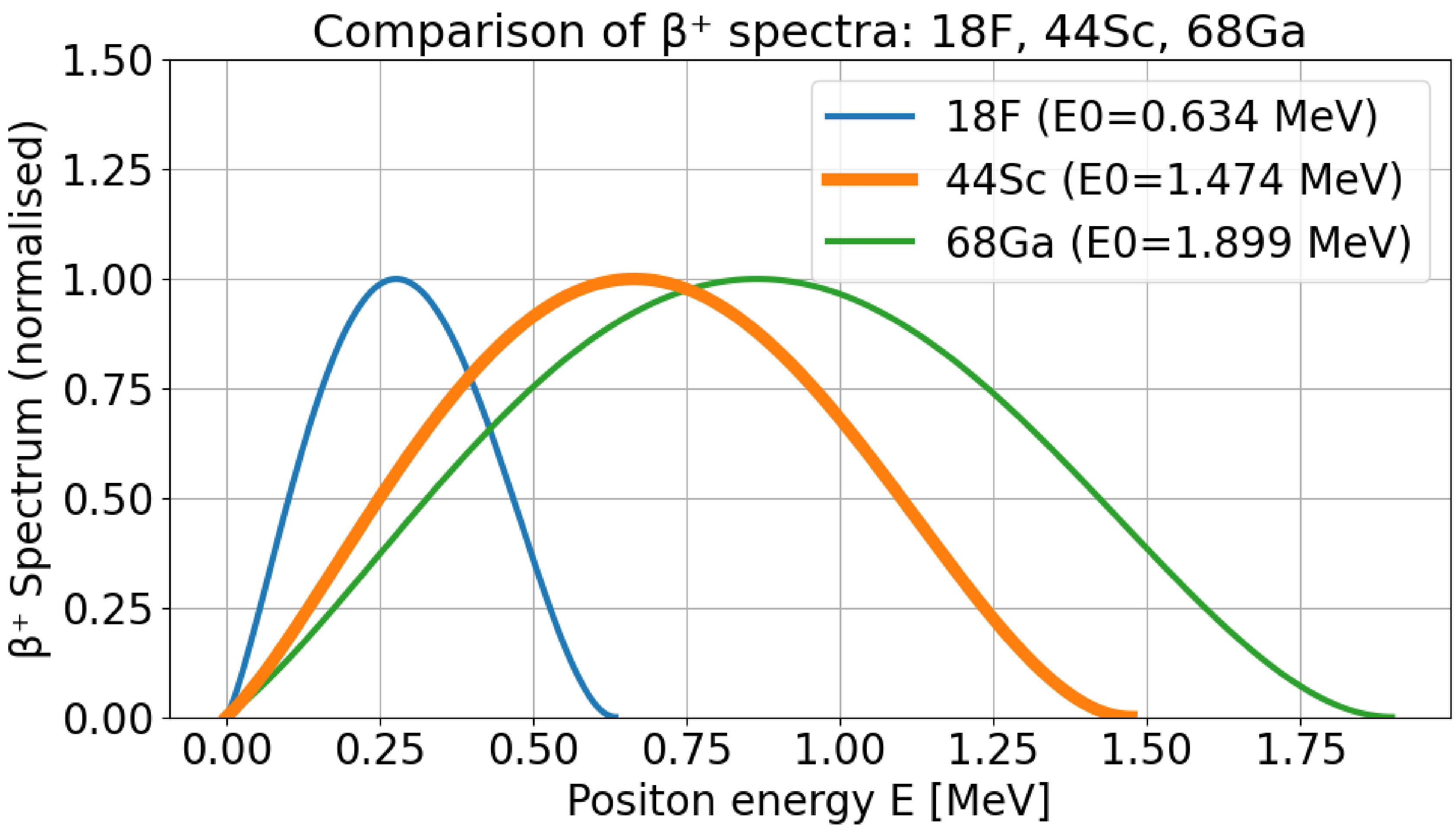
Carter: https://pmc.ncbi.nlm.nih.gov/articles/PMC6805144/#TFN1

Table 4. Positron ranges in water for each radioisotope studied. Comparison between our Monte Carlo results (in bold) and existing values taken from different sources: for R_{mean}: Partridge et al (2006) (reference (1)), Bailey et al (2003) (reference (2)), Comtat (private communication) (reference (3)) and Defrise and Trebossen (private communication) (reference (4)); for R_{max}: Evans (1972) (reference (5)). Experimental measurements are taken from Derenzo (1993) and Derenzo et al (1930) (reference (6)) and from Cho et al (1975) (reference (7)). For all these values, relative disagreement with our result is reported into brackets.

	R_{mean} (mm)		R_{max} (mm)
Isotopes	Theory	Experimental	Theory
^{18}F	**0.66** 0.6[1] (9.1%) 0.6[2] (9.1%) 0.5437[3] (17.6%) 0.6[4] (9.1%)	0.54[6] (18.2%) 0.9[7] (36.1%)	**2.633** 2.3[5] (12.6%)
^{11}C	**1.266** 1.1[1] (13.1%) 1.1[2] (13.1%) 1.12[4] (11.5%)	0.92[6] (27.3%) 1.095[7] (13.5%)	**4.456** 3.9[5] (12.5%)
^{13}N	**1.730** 1.5[1] (13.3%) 1.5[2] (13.3%) 1.44[4] (16.8%)	1.39[7] (19.6%)	**5.752** 5.1[5] (11.3%)
^{15}O	**2.965** 2.5[1] (15.7%) 2.5[2] (15.7%) 2.261[3] (23.8%) 2.22[4] (25.1%)	1.785[7] (39.8%)	**9.132** 8[5] (12.4%)
^{68}Ga	**3.559** 2.9[1] (18.5%) 2.4[4] 32.6%)	2.8[6] (21.3%) 1.975[7] (44.5%)	**10.273** 9[6] (12.4%)
^{82}Rb	**7.491** 5.9[1] (21.2%) 4.7[4] (37.2%)	6.1[6] (18.6%) 2.9[7] (61.2%)	**18.603** 18[5] (3.2%)

Champion: https://iopscience.iop.org/article/10.1088/0031-9155/52/22/004/pdf

Energy spectrum



$$N(E)\,dE = g\,F(Z,E)\,pE(E_{\max}-E)^2\,dE$$

$$F_{\text{allowed}}(Z,E) = \frac{2\pi\eta}{1-e^{-2\pi\eta}}$$

Levin CS, Hoffman EJ. Calculation of positron range and its effect on PET resolution. Phys Med Biol.

Table with 44Sc

Radionuclide	Energy_mean [MeV]	Range_Lit [mm]	Range_calulated [mm]	Difference in Range
Mn-52	0,242	0,59	0,565	-4,24%
F-18	**0,25**	**0,6**	**0,594**	**-1,03%**
Cu-64	0,28	0,7	0,705	0,77%
C-11	0,39	1,1	1,150	4,58%
Zr-89	0,4	1,1	1,193	8,47%
N-13	0,49	1,5	1,592	6,13%
Sc-44	**0,63**	**2**	**2,250**	**12,51%**
Y-86	0,66	2,1	2,396	14,09%
O-15	0,74	2,4	2,791	16,28%
I-129	0,82	2,9	3,193	10,11%
Ga-68	**0,83**	**2,8**	**3,244**	**15,86%**

$$r = 0.412 \cdot E^{(1.265 - 0.0954 \cdot \ln E)}$$

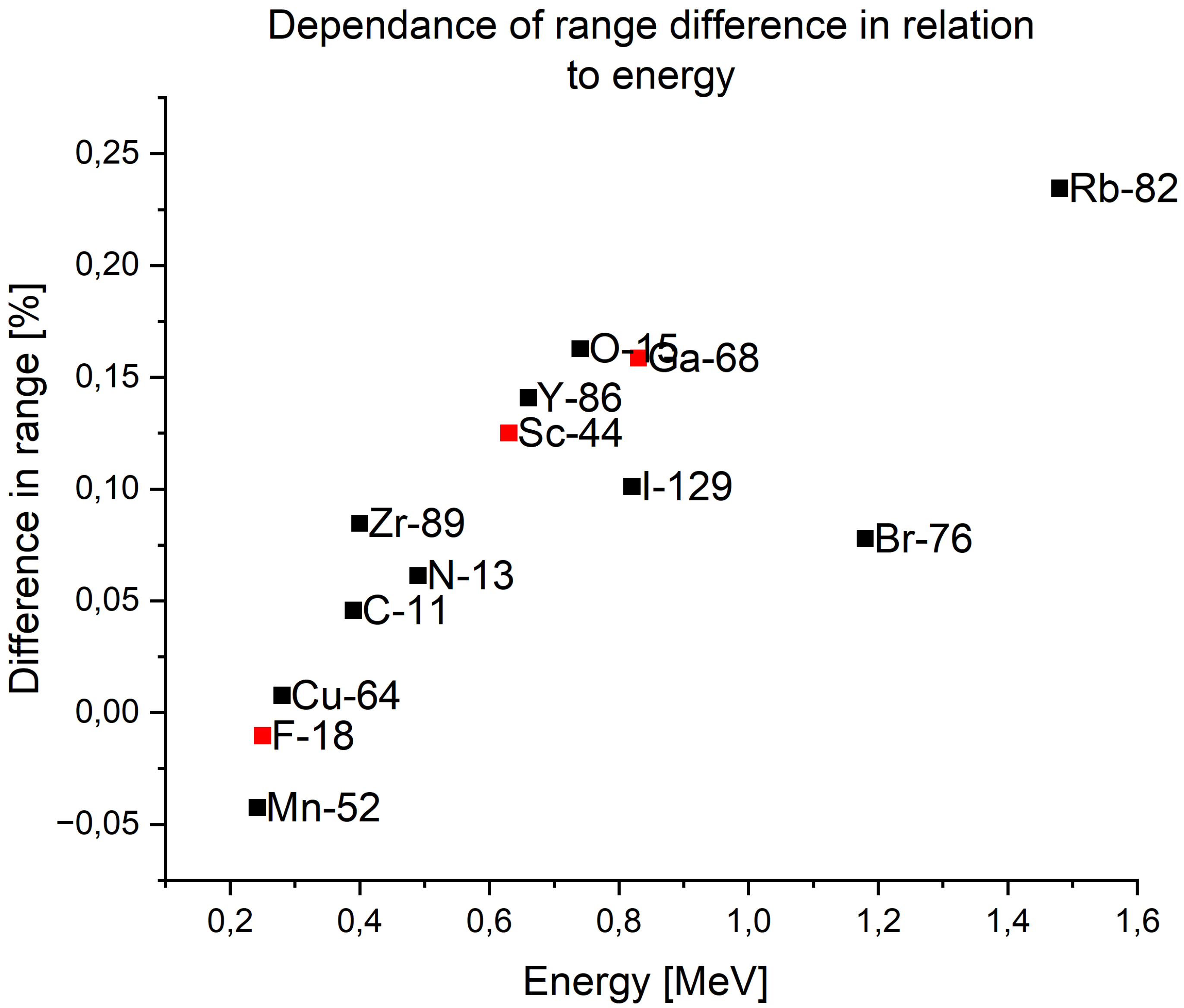
$$X = \frac{r}{\rho}$$

$$\text{Difference in range} = \frac{\text{Range\_calculated} - \text{Range\_Lit}}{\text{Range\_Lit}}$$

https://bamsjournal.com/article/541939/en

https://www.nupecc.org/pub/npmed2014.pdf

Range Difference vs. Energy



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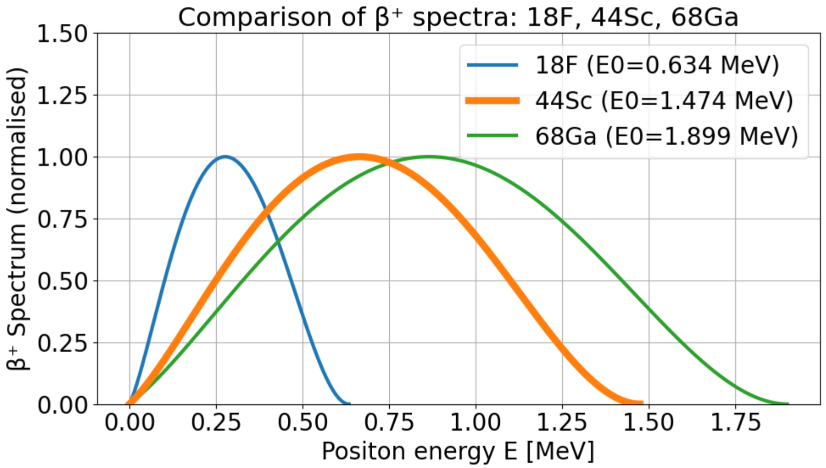
Holistic approach



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$$r = 0.412 \cdot E^{(1.265 - 0.0954 \cdot \ln E)}$$

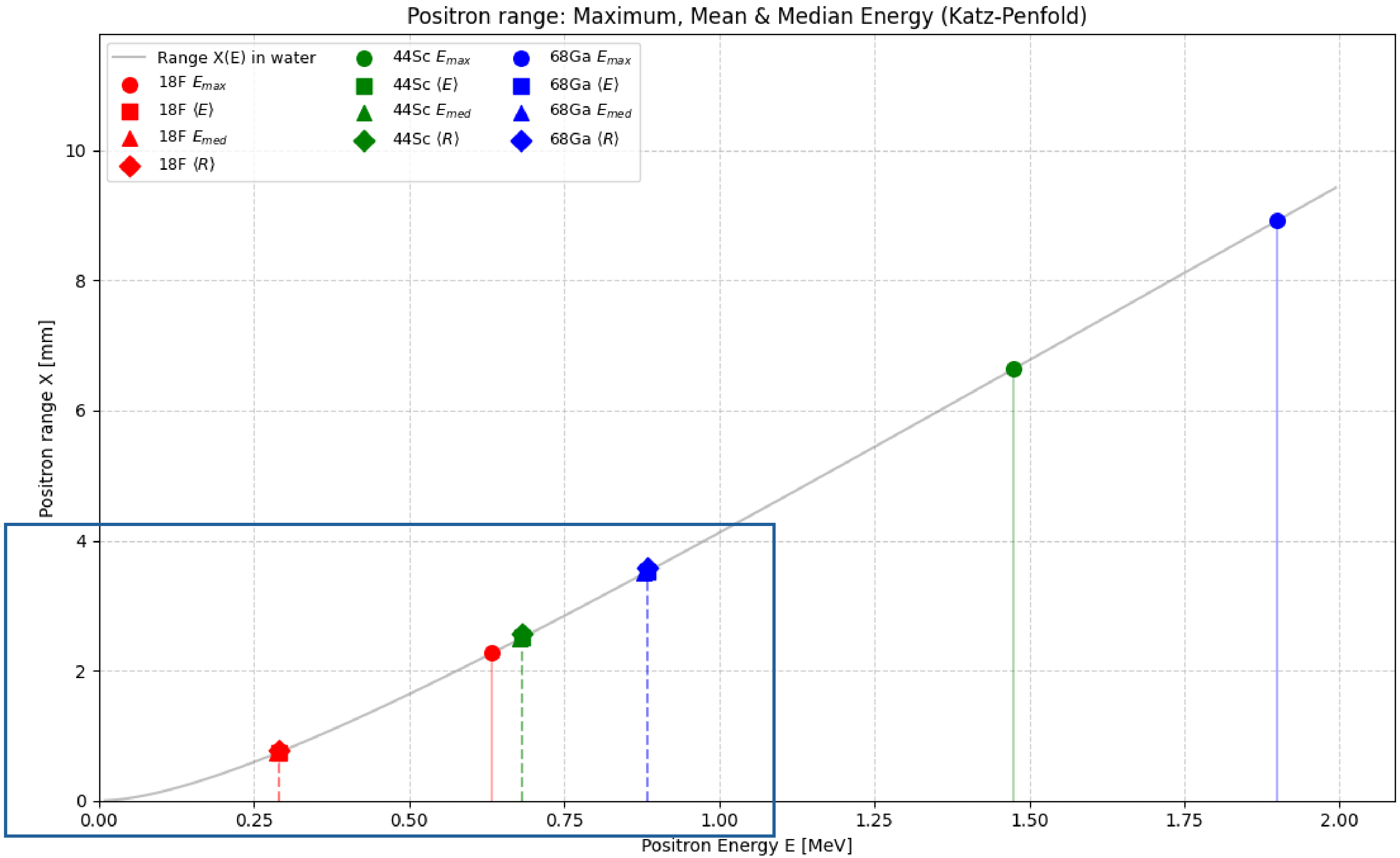
$$X = \frac{r}{\rho}$$

Compute X over the whole spectrum and see the results

https://bamsjournal.com/article/541939/en

Levin CS, Hoffman EJ. Calculation of positron range and its effect on PET resolution. Phys Med Biol.

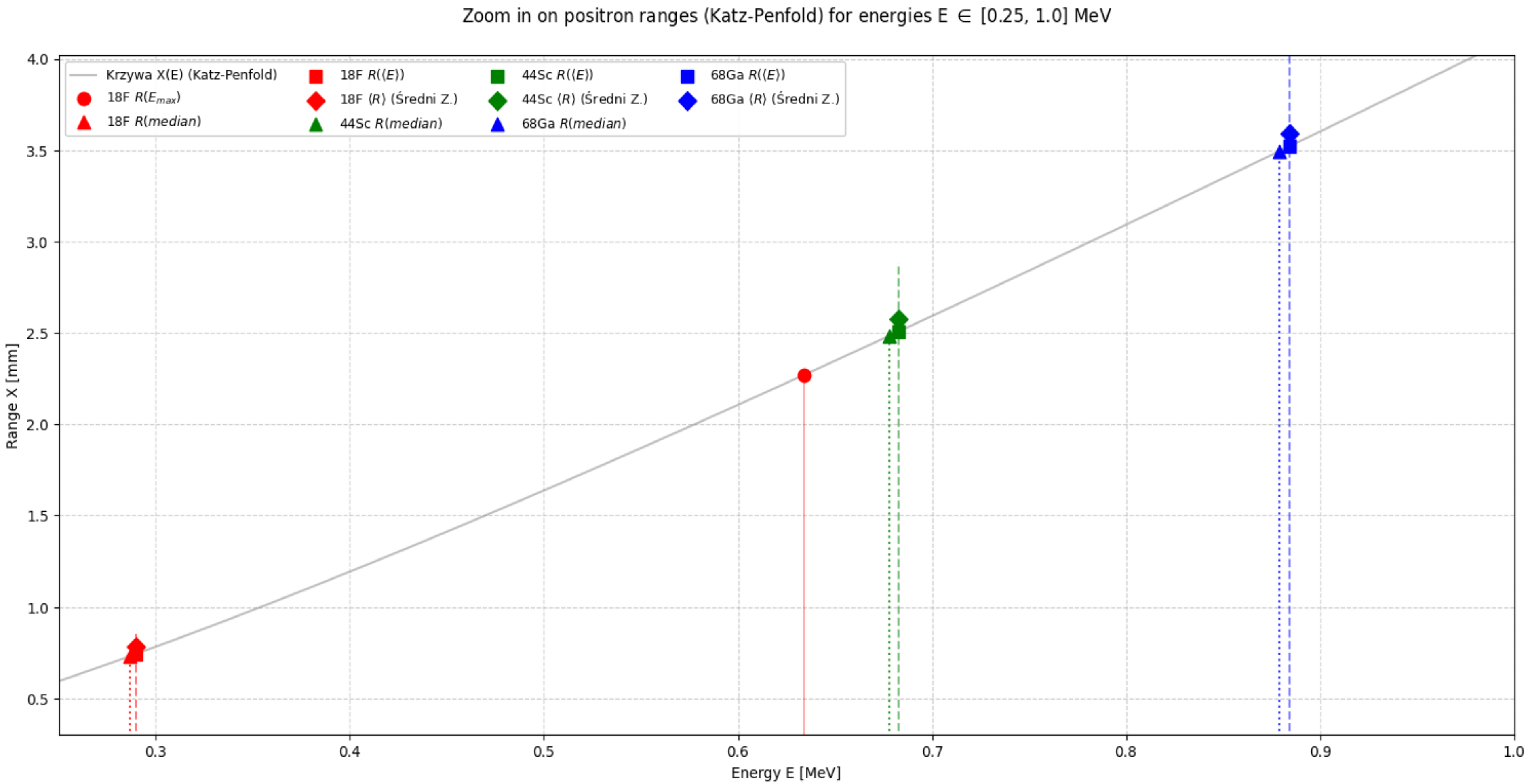
Holistic approach



$$r = 0.412 \cdot E^{(1.265 - 0.0954 \cdot \ln E)} \qquad X = \frac{r}{\rho}$$

https://bamsjournal.com/article/541939/en

Zoom in

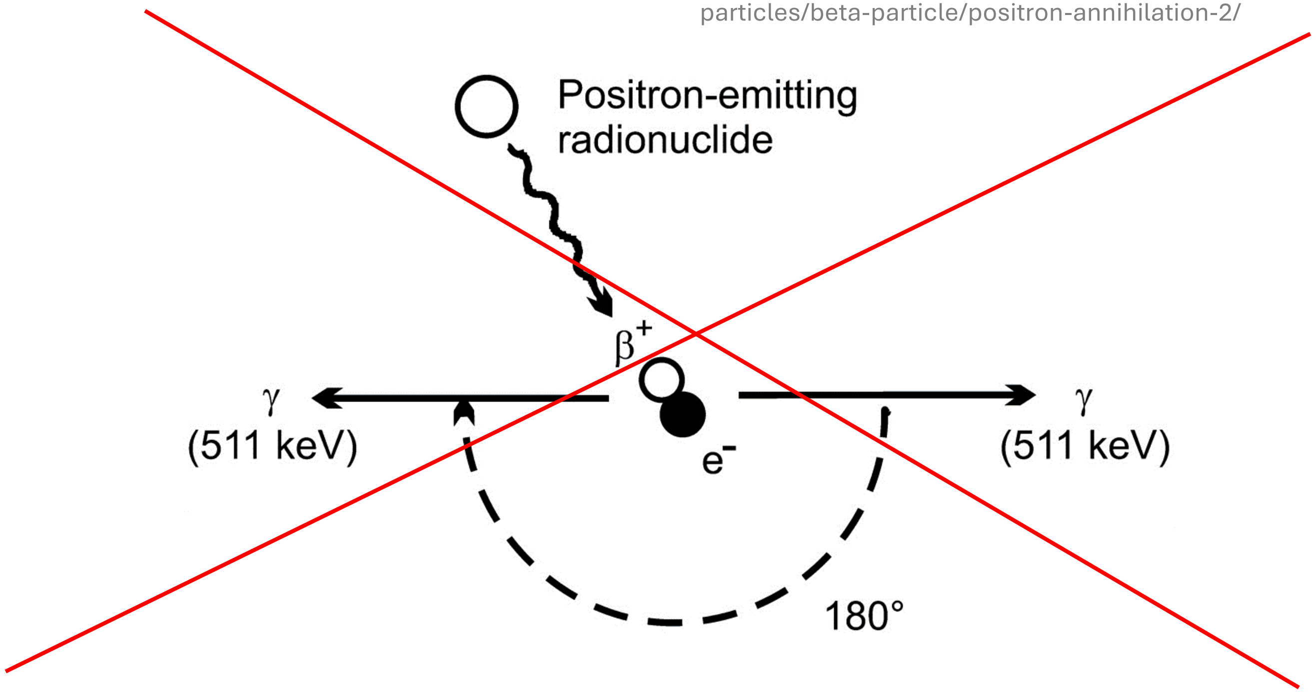


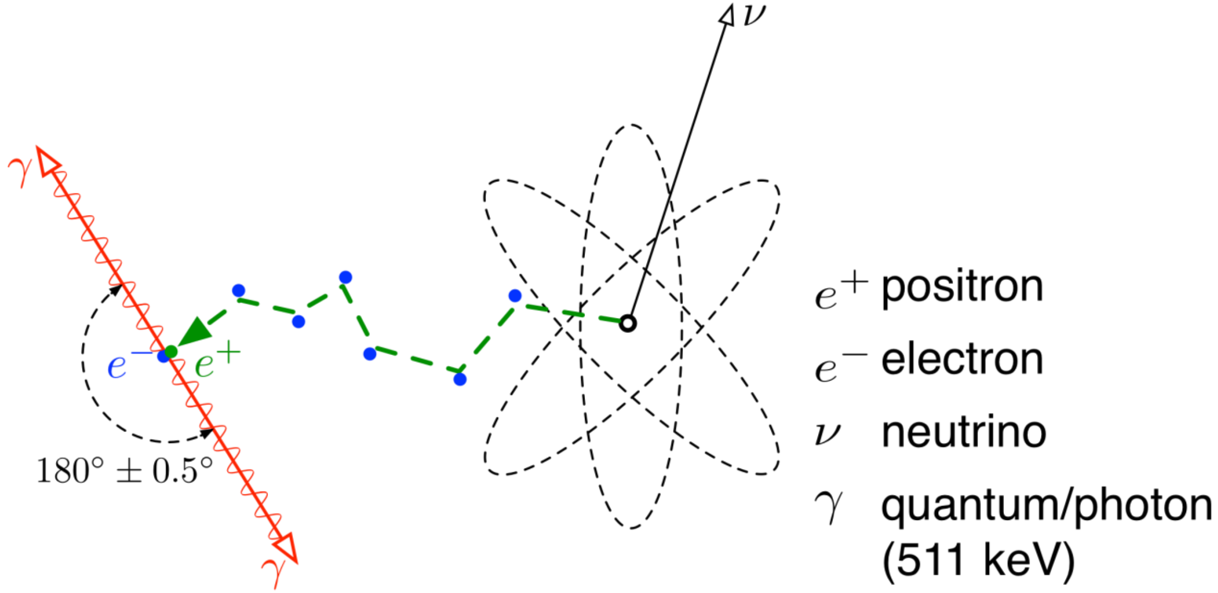
Isotope	R_max [mm]	R_mean (<E>) [mm]	<R_mean> (true) [mm]	R_median [mm]
18F	2,270	0,742	0,783	0,730
44Sc	6,635	2,509	2,577	2,485
68Ga	8,916	3,521	3,591	3,494

https://bamsjournal.com/article/541939/en

The positron path

https://www.nuclear-power.com/nuclear-power/reactor-physics/atomic-nuclear-physics/fundamental-particles/beta-particle/positron-annihilation-2/

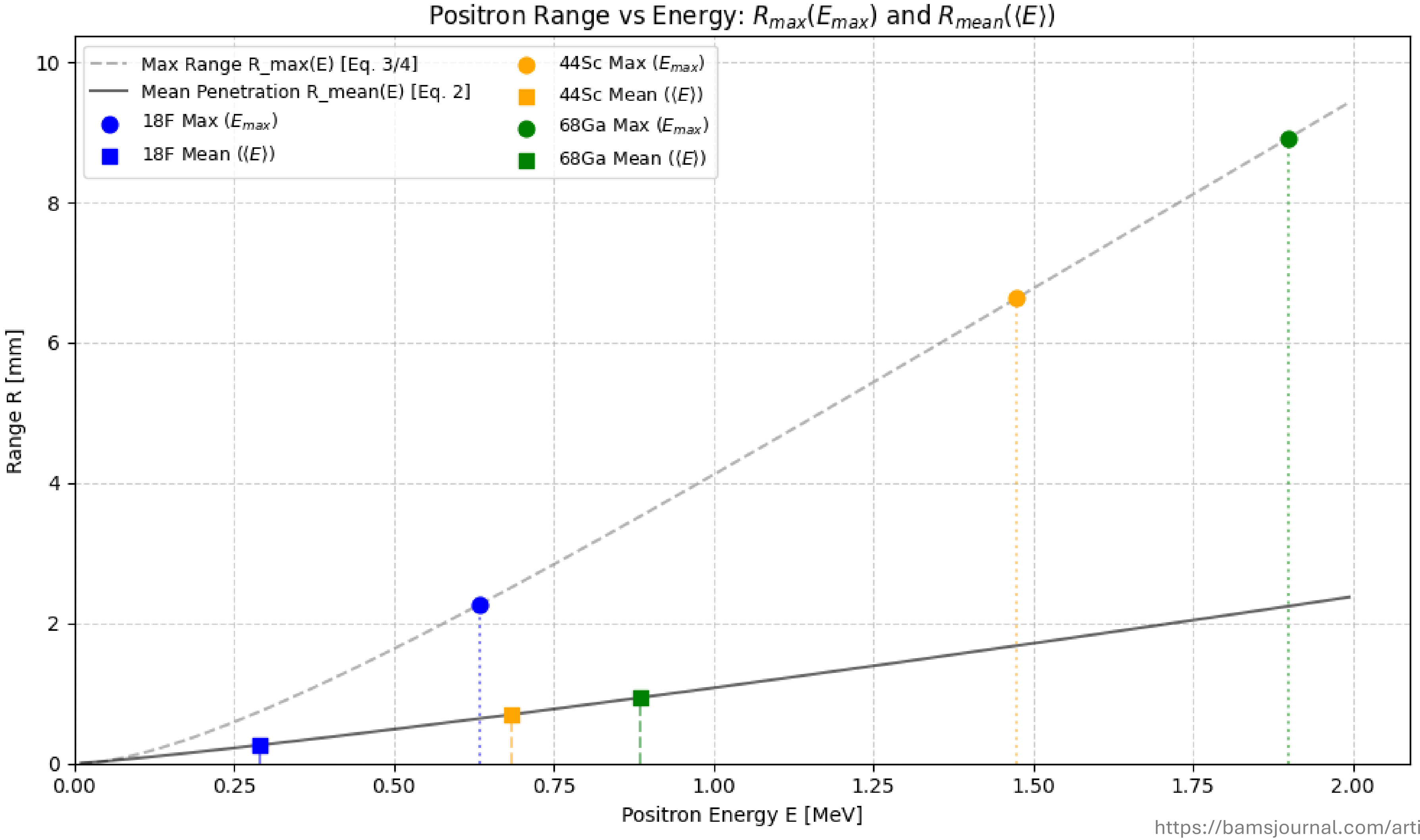




https://commons.wikimedia.org/wiki/File:Annihilation.png

The positron range

$$R_{mean} \approx \frac{0.108(E_{e^+}(MeV))^{1.14}}{\rho(g{\cdot}cm^{-3})}$$



https://bamsjournal.com/article/541939/en

The positron range

Isotope	R_max [mm]	R_mean (<E>) [mm]	<R_mean> (true) [mm]	R_median [mm]
18F	2,270	0,263	0,267	0,260
44Sc	6,635	0,699	0,710	0,694
68Ga	8,916	0,939	0,953	0,932

$$R_{mean} \approx \frac{0.108(E_{e^+}(MeV))^{1.14}}{\rho(g{\cdot}cm^{-3})}$$

https://bamsjournal.com/article/541939/en

Theorethical spheres

Isotope	R_max [mm]	R_mean (<E>) [mm]	<R_mean> (true) [mm]	R_median [mm]
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44Sc	6,635	2,509	2,577	2,485
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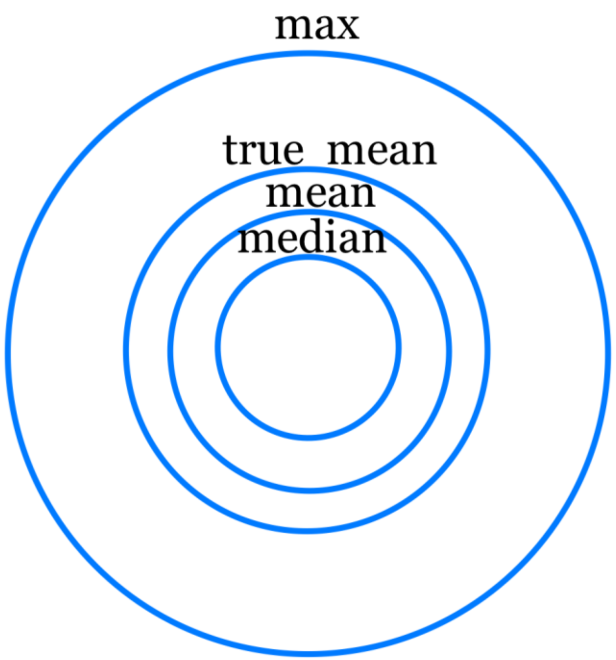
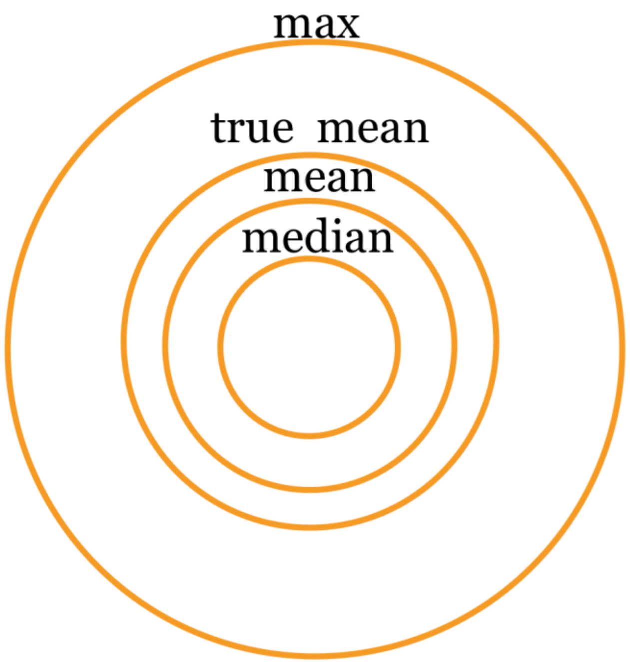
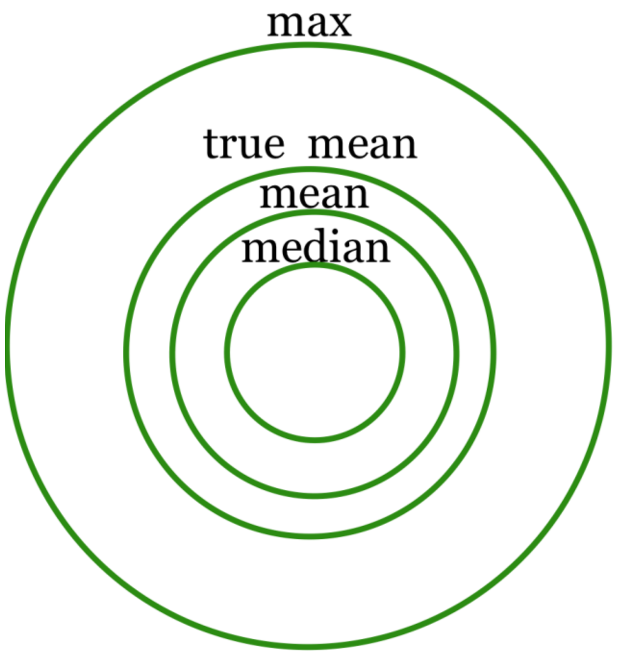
18F



44Sc



68Ga



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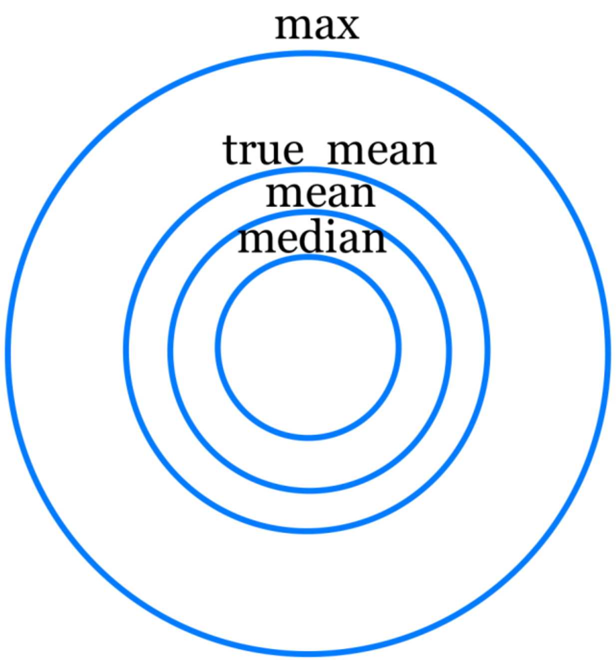
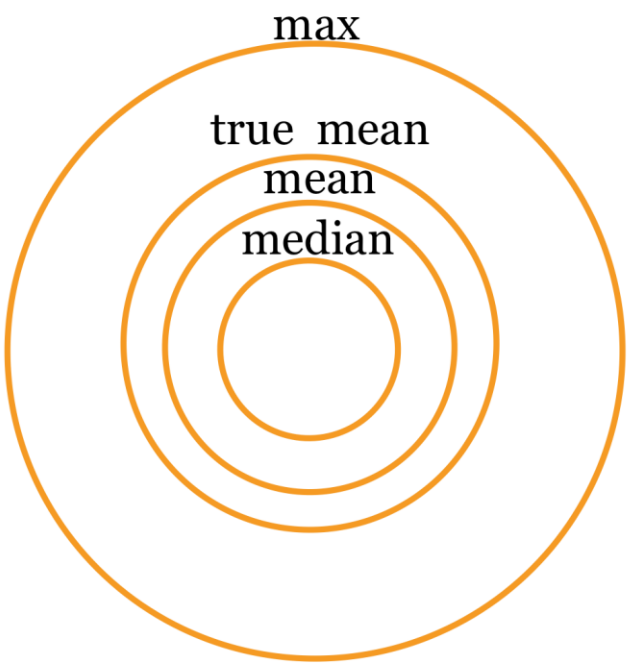
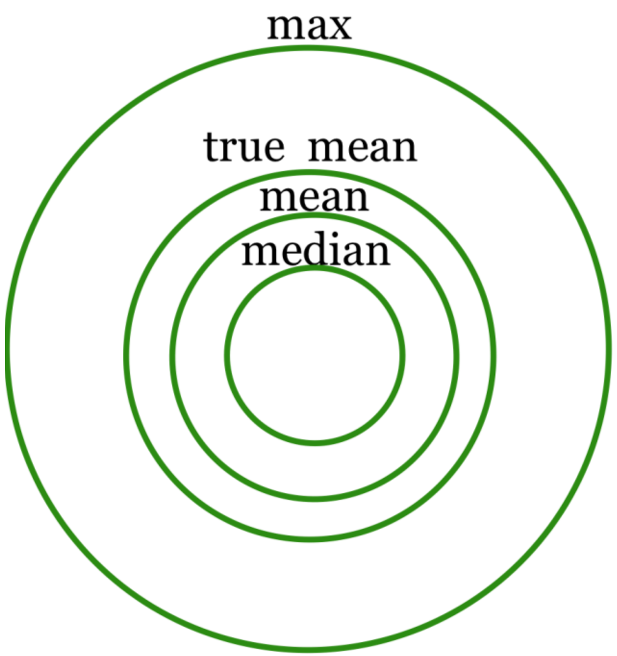
18F



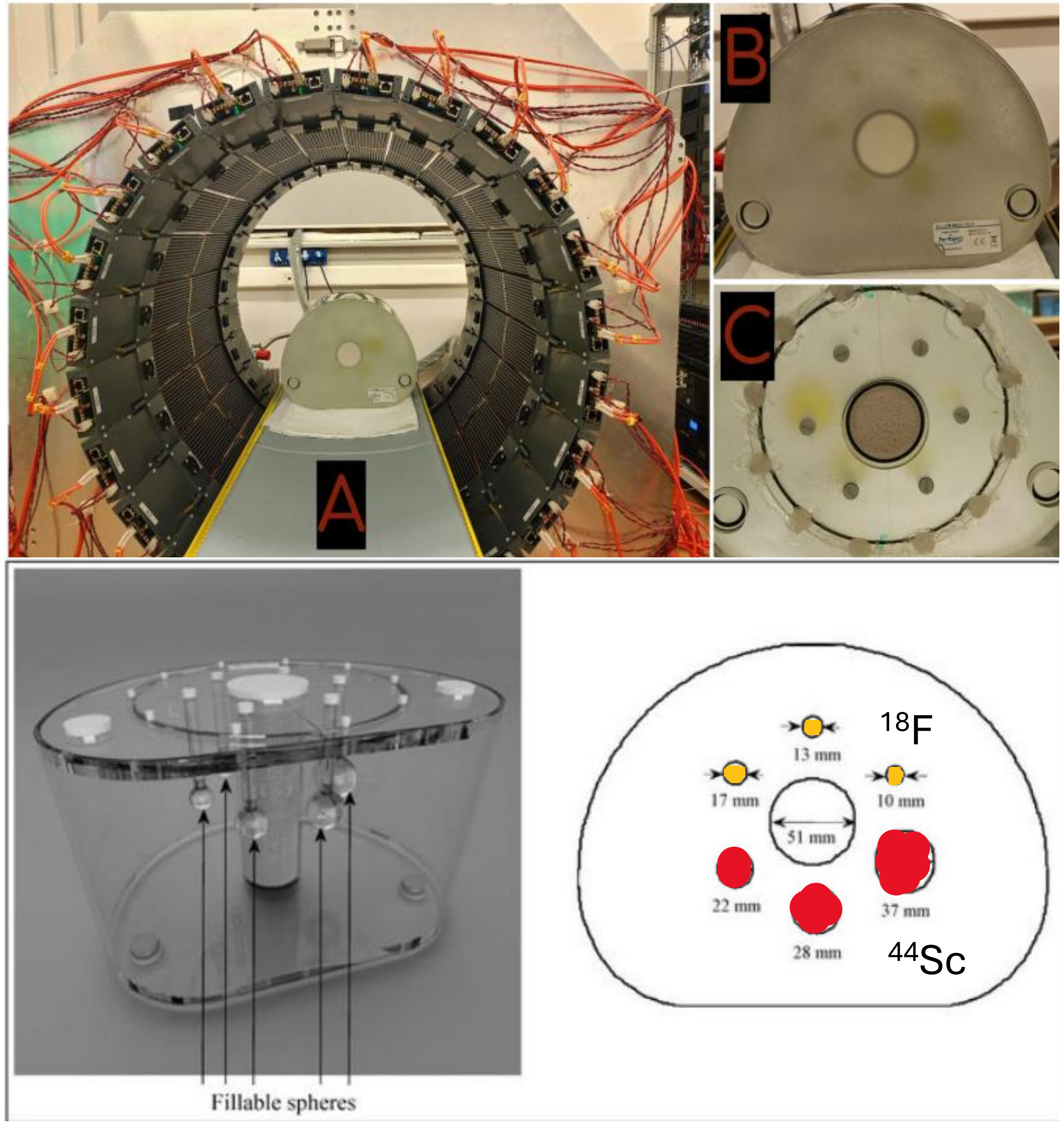
44Sc



68Ga



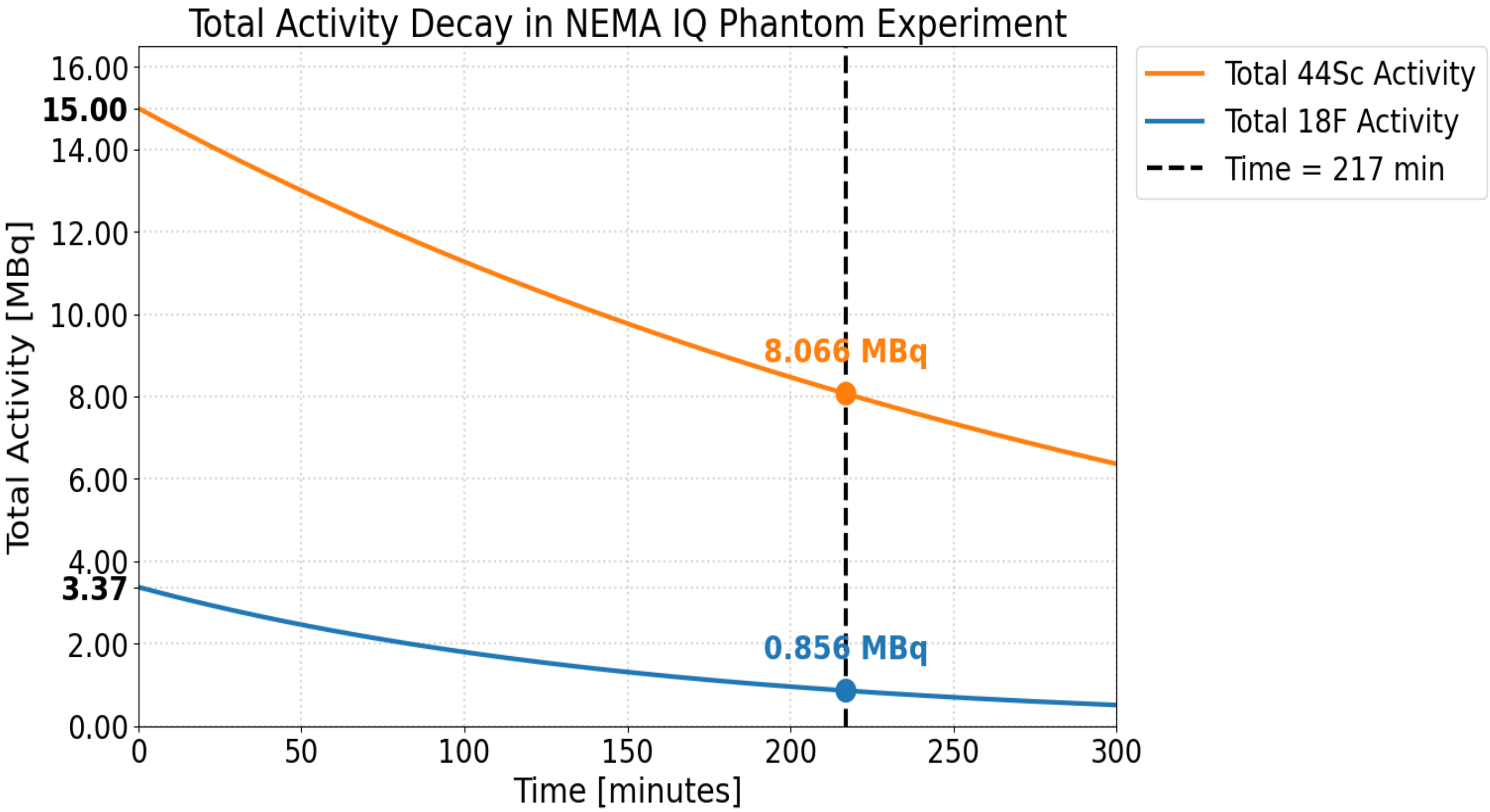
Experiment



Raport_Scandium_exp2025
section 3.4.4

Activity decay

$^{44}Sc/^{18}F$ =
4,45104



$^{44}Sc/^{18}F$ =
9,42290

$$A = A_0 e^{-\lambda t} \quad \lambda = \frac{\ln(2)}{T_{1/2}}$$

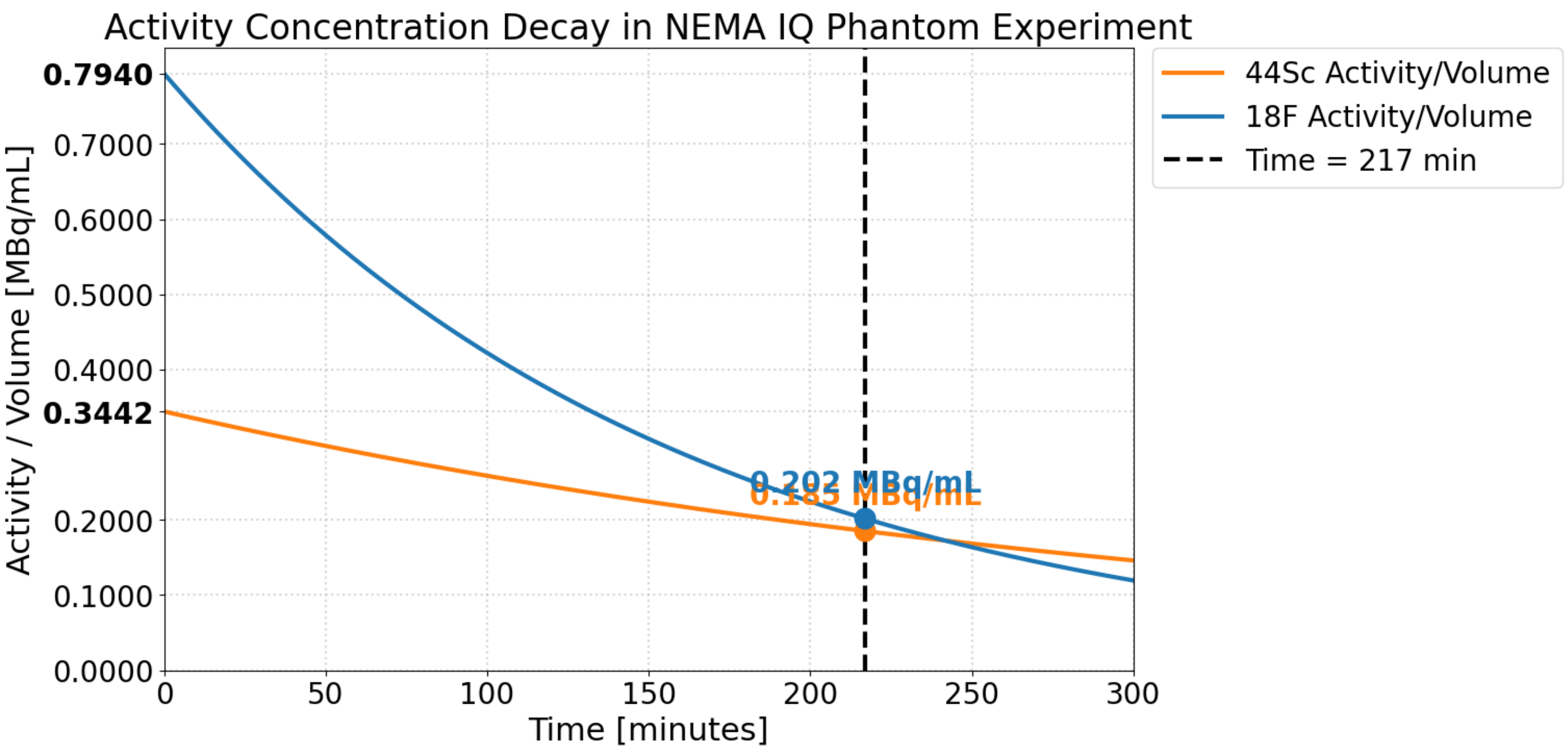
$$T_{1/2}^{Sc} = 242.52\,min$$

$$T_{1/2}^{F} = 109.73\,min$$

doi.org/10.1016/j.apradiso.2022.110507

Activity decay

$^{44}Sc/^{18}F$ = 0,4335



$^{44}Sc/^{18}F$ = 0,7088

$$\frac{A}{V} = \frac{A_0}{V} e^{-\lambda t} \qquad \lambda = \frac{\ln(2)}{T_{1/2}}$$

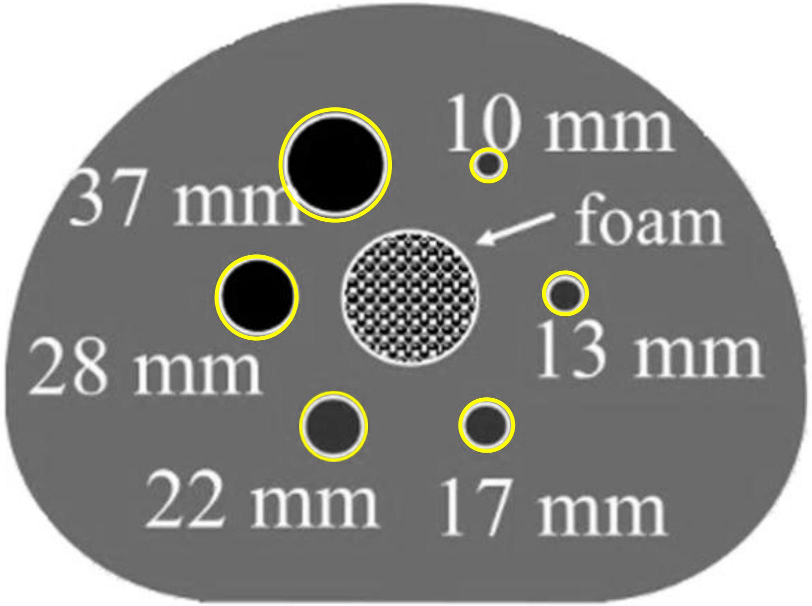
$$T_{1/2}^{Sc} = 242.52\, min$$

$$T_{1/2}^{F} = 109.73\, min$$

doi.org/10.1016/j.apradiso.2022.110507

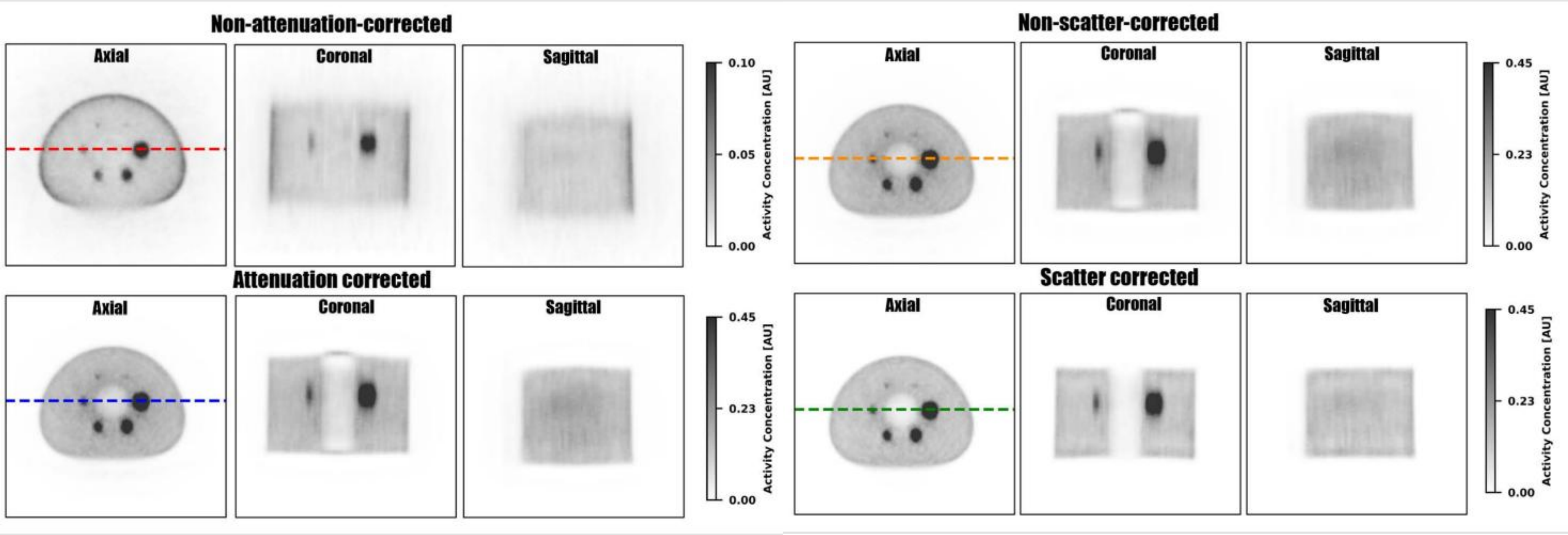
Image reconstruction

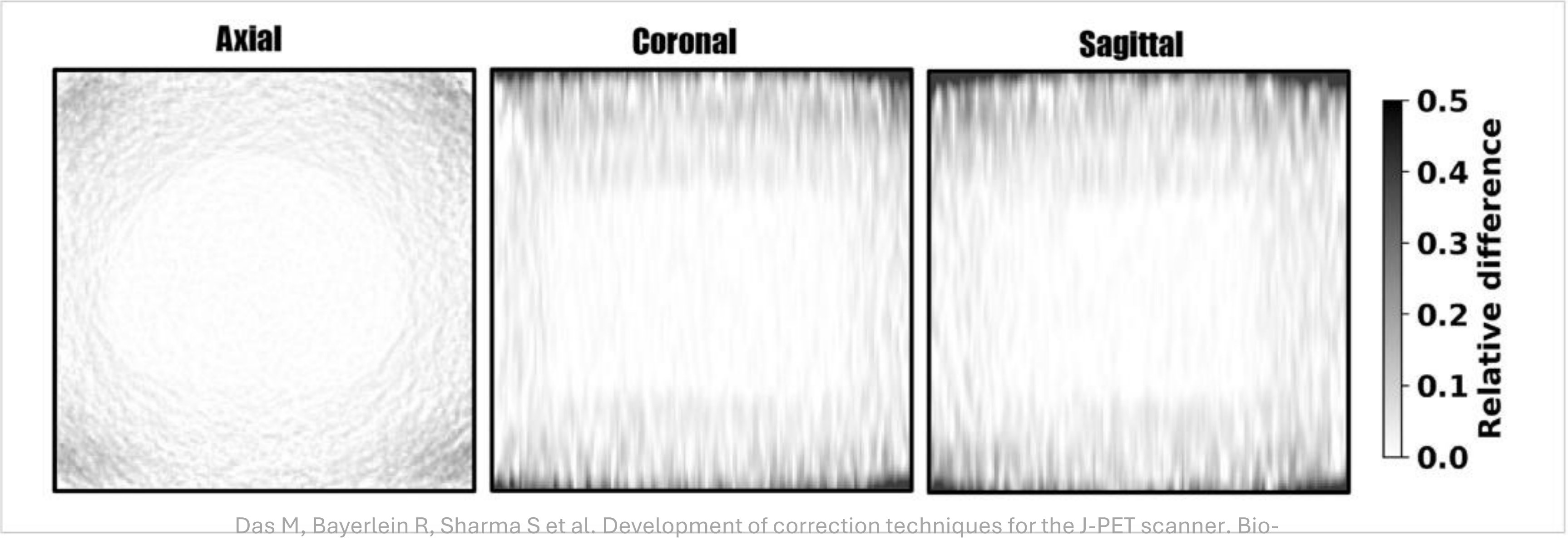
Data from the experiment



https://link.springer.com/article/10.1186/s40658-020-00294-y/figures/1

Applying corrections





Das M, Bayerlein R, Sharma S et al. Development of correction techniques for the J-PET scanner. Bio-Algorithms and Med-Systems. (2024);20(1):101-110. https://doi.org/10.5604/01.3001.0054.9362.

The Aim

Literature

vs

Analytical

vs

Experiment

vs (?)

MC

The Questions

1. Theorethical ranges based on emission spectra were calculated assuming a point source. How to approach the matter when in reality spheres were used?

2. How to differentiate the median range vs mean range vs true mean range on the reconstructed image?

Thank You for Your attention

Back up

1. TOT 3-5ns
2. hit multiplicity 2
3. angular separation (2dim) - 60<=
4. geometrical -23<=z=<23
5. TOF <= 2.5ns

8 coordinates -> 1 event. Export it as a LUT file into castor

1. dim: 200*200*200
2. voxel = 2.5mm
3. MLEM/OSEM (mlem better but osem good for a lot of statistics)
4. 10 iterations
5. OSEM: interation/subste check

Gaussian smoothing sigma = 2voxels

Back up